

Algebraic structures of the Lindblad equation

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Abstract

This Mathematica notebook accompanies the paper The Algebraic Structure of the Lindblad Equation. The notebook illustrates the proposed algebraic formulation through the example of a driven transmon qubit described by the Lindblad master equation.

The example is organized as a step-by-step implementation of the algebraic method. It begins by constructing the Hermitian matrix basis and the associated algebraic objects, followed by the recursive generation of the operators required to represent the Liouville superoperator. The model-dependent quantities are then assembled to obtain the complete Liouville superoperator, which is compared with the one obtained by the conventional direct construction, demonstrating their exact agreement.

The notebook is fully parameterized with respect to the number of qubits, allowing the same implementation to be extended to larger Hilbert spaces. Although some symbolic calculations become computationally demanding as the system size increases, the code provides a general framework that can be readily adapted to study a broad class of finite-dimensional open quantum systems.

Licence

LindbladAlgebra Mathematica examples accompanying the paper

“The Algebraic Structure of the Lindblad Equation”

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Example

Functions

One-qubit matrix basis $\mathbf{h}$

In the following example, we consider a single qubit, although the number of qubits can be increased. Some of the calculations, however, involve high-rank tensors that, for systems with more than two qubits, may exceed the available memory. Whenever this limitation arises, it is explicitly indicated in the text.

```
In[ ]:= nq = 1;
```

We begin by constructing the one-qubit basis $\mathbf{h}_n$ and its associated structure constant superoperators $\mathbf{Z}$, $\mathbf{Z}^*$ and $\mathbf{X}$.

```
In[ ]:= h = QubitBasis[nq];
{b, c} = BCStructureConstantsFromQubits[nq];
z = ZStructureConstantsFromQubits[nq];
zt = Conjugate[z];
x = XBasisFromQubits[nq];
```

The complex structure-constant superoperators can be visualized with the following line of code.

```
In[ ]:= MatrixForm[z[[3]]]
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 & -i\sqrt{2} \\ \sqrt{2} & 0 & 0 & 0 \\ 0 & i\sqrt{2} & 0 & 0 \end{pmatrix}$$

It can be verified that $\mathbf{Z}$ and $\mathbf{Z}^*$ constitute to independent realizations of the Lie algebra associated with $\mathbf{h}$ by showing that the following difference, obtained by subtracting the right-hand side from the left-hand side of Eqs. (65) and (66), vanish identically.

```
In[*]:= zz = Transpose[z.Transpose[z], {1, 3, 2, 4}];
commu = SparseArray[zz - Transpose[zz]];
PossibleZeroQ[SparseArray[commu + 2 I c.z]]
```

Out[*]=

SparseArray[ Specified elements: 0
Dimensions: {4, 4, 4, 4}
Default: True]

```
In[*]:= ztzt = Transpose[zt.Transpose[zt], {1, 3, 2, 4}];
commu = SparseArray[ztzt - Transpose[ztzt]];
PossibleZeroQ[SparseArray[commu - 2 I c.zt]]
```

Out[*]=

SparseArray[ Specified elements: 0
Dimensions: {4, 4, 4, 4}
Default: True]

Similarly we can verify that $\mathbf{Z}, \mathbf{Z}^*$ commute.

```
In[*]:= zzt = Transpose[z.ConjugateTranspose[z]];
ztz = Transpose[zt.Transpose[z], {3, 1, 2, 4}];
commu = SparseArray[zzt - ztz];
PossibleZeroQ[commu]
```

Out[*]=

SparseArray[ Specified elements: 0
Dimensions: {4, 4, 4, 4}
Default: True]

The structure of the 16 elements of $\mathbf{X}$ can be visualized using the following line of code.

```
In[*]:= MatrixForm[x[[2, 3]]]
```

Out[*]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 \end{pmatrix}$$

The Lie and Jordan algebraic relations of $\mathbf{X}$ can be verified by evaluating the following differences, obtained by subtracting the right-hand side from the left-hand side of Eqs. (72) and (74), and verifying that they vanish identically. This calculation may exceed the available memory if $n_q \geq 2$.

```
In[*]:= xx = Transpose[x.Transpose[x, {2, 3, 1, 4}], {1, 2, 5, 3, 4, 6}];
commu = SparseArray[xx - Transpose[xx, {3, 4, 1, 2, 5, 6}]];
anticomu = SparseArray[xx + Transpose[xx, {3, 4, 1, 2, 5, 6}]];
```

```
In[*]:= PossibleZeroQ[SparseArray[commu -
  
$$\frac{1}{4} \text{Transpose}[(\text{zt}.\text{Transpose}[x, \{1, 4, 2, 3\}].z - z.\text{Transpose}[x, \{1, 4, 2, 3\}].\text{zt}),$$

  {1, 3, 5, 6, 2, 4}]]]
```

Out[*]=

SparseArray[ Specified elements: 0
Dimensions: {4, 4, 4, 4, 4, 4}
Default: True]

```
In[*]:= PossibleZeroQ[SparseArray[anticommu -
  
$$\frac{1}{4} \text{Transpose}[(\text{zt}.\text{Transpose}[x, \{1, 4, 2, 3\}].z + z.\text{Transpose}[x, \{1, 4, 2, 3\}].\text{zt}),$$

  {1, 3, 5, 6, 2, 4}]]]
```

Out[*]=

SparseArray[ Specified elements: 0
Dimensions: {4, 4, 4, 4, 4, 4}
Default: True]

The orthonormality of X_{n^2} is verified by flattening the first two indices of X into a single index and then evaluating the Hilbert-Schmidt inner product

```
In[*]:= X = Flatten[x, 1]
```

Out[*]=

SparseArray[ Specified elements: 64
Dimensions: {16, 4, 4}]

```
In[*]:= XX = SparseArray[TensorContract[X.Transpose[X], {{2, 4}}]]
DiagonalMatrixQ[XX]
```

Out[*]=

SparseArray[ Specified elements: 16
Dimensions: {16, 16}]

Out[*]=

True

Hamiltonian

A convenient system for illustrating the results developed in this work is the transmon qubit. Since the number of quantum levels retained in the model can be chosen arbitrarily, it provides a simple framework for studying the behavior of the algebraic structures introduced above as the Hilbert-space dimension increases. We first define the lowering and raising operators as well as the jump operators

```

In[*]:= m = 2nq;
n = 22 nq;
a = SparseArray[Table[{n + 1, n + 2} →  $\sqrt{n + 1}$ , {n, 0, m - 2}], m];
at = SparseArray[Table[{n + 1, n} →  $\sqrt{n}$ , {n, 1, m - 1}], m];
A = SparseArray[Table[{n + 1, n + 1, n + 2} →  $\sqrt{n + 1}$ , {n, 0, m - 2}], {m - 1, m, m}];
At = SparseArray[Table[{n, n + 1, n} → 1, {n, 1, m - 1}], {m - 1, m, m}];

```

The Hamiltonian in the interaction picture, given by Eq. (102) is given by

```

In[*]:= Hi =  $\frac{\Omega}{2}$  (a Exp[i Δ t] + at Exp[-i Δ t]);
Hi // MatrixForm

```

```

Out[*]//MatrixForm=

$$\begin{pmatrix} 0 & \frac{1}{2} e^{i t \Delta \Omega} \\ \frac{1}{2} e^{-i t \Delta \Omega} & 0 \end{pmatrix}$$


```

The Hamiltonian vector is obtained by projecting the Hamiltonian on the elements of the set $\mathbf{h}_n$ through Eq. (21).

```

In[*]:= H = ExpToTrig[ProjectOperatorOntoBasis[Hi, h]];
MatrixForm[H]

```

```

Out[*]//MatrixForm=

$$\begin{pmatrix} 0 \\ \frac{\Omega \cos[t \Delta]}{\sqrt{2}} \\ -\frac{\Omega \sin[t \Delta]}{\sqrt{2}} \\ 0 \end{pmatrix}$$


```

Lindbladian

The Lindbladian part of the Liouville operator requires Γ , which is defined in Eq. (36).

```

In[*]:= γ = SparseArray[Table[{ii, ii} → γ0, {ii, m - 1}], {m - 1, m - 1}];
Γ = GammaFromgamma[h, γ, A]
MatrixForm[Γ]

```

```

Out[*]=
SparseArray[ Specified elements: 4  
Dimensions: {4, 4}]

```

```

Out[*]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{\gamma_0}{2} & \frac{i \gamma_0}{2} & 0 \\ 0 & -\frac{i \gamma_0}{2} & \frac{\gamma_0}{2} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$


```

Algebraic Method

Having obtained the quantities required to construct the Hamiltonian and Lindbladian contributions to the Liouville superoperator, namely $\mathbf{H}$ and $\mathbf{\Gamma}$, we can proceed to compute Λ using Eq. (55), yielding

```
In[ ]:=  $\Lambda$  = LambdaFromHamiltonianLindbladian[z, H,  $\Gamma$ ]
MatrixForm[ $\Lambda$ ]
```

```
Out[ ]:=
```

```
SparseArray[  Specified elements: 11  
Dimensions: {4, 4} ]
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} -\gamma^0 & i\Omega \cos[t\Delta] & -i\Omega \sin[t\Delta] & \frac{\gamma^0}{2} \\ -i\Omega \cos[t\Delta] & \frac{\gamma^0}{2} & \frac{i\gamma^0}{2} & 0 \\ i\Omega \sin[t\Delta] & -\frac{i\gamma^0}{2} & \frac{\gamma^0}{2} & 0 \\ \frac{\gamma^0}{2} & 0 & 0 & 0 \end{pmatrix}$$

The Liouville superoperator is obtained using Eq. (54).

```
In[ ]:=  $\mathbf{Lam}$  = LiouvilleS0FromLamda[x,  $\Lambda$ ]
MatrixForm[ $\mathbf{Lam}$ ]
```

```
Out[ ]:=
```

```
SparseArray[  Specified elements: 16  
Dimensions: {4, 4} ]
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -\frac{\gamma^0}{2} & 0 & \Omega \sin[t\Delta] \\ 0 & 0 & -\frac{\gamma^0}{2} & \Omega \cos[t\Delta] \\ \gamma^0 & -\Omega \sin[t\Delta] & -\Omega \cos[t\Delta] & -\gamma^0 \end{pmatrix}$$

Direct method

The Liouville superoperator is also computed in the same matrix basis for comparison. The two methods are found to yield identical results.

```
In[ ]:=  $\mathbf{Ldm}$  = LiouvilleS0FromProjection[h, H,  $\Gamma$ ]
MatrixForm[ $\mathbf{Ldm}$ ]
```

```
Out[ ]:=
```

```
SparseArray[  Specified elements: 16  
Dimensions: {4, 4} ]
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -\frac{\gamma^0}{2} & 0 & \Omega \sin[t\Delta] \\ 0 & 0 & -\frac{\gamma^0}{2} & \Omega \cos[t\Delta] \\ \gamma^0 & -\Omega \sin[t\Delta] & -\Omega \cos[t\Delta] & -\gamma^0 \end{pmatrix}$$

Comparison of the algebraic method and the direct method

To compare the Liouville superoperators obtained using the algebraic and direct methods, we subtract one superoperator from the other.

```
In[*]:= PossibleZeroQ[SparseArray[Lam - Ldm]]
Out[*]=
```

```
SparseArray[ Specified elements: 0  
Dimensions: {4, 4}  
Default: True]
```

Liouville equation for the dynamical map

To verify that the dynamical differential equations for the coordinate vector, Eq. (97), are equivalent to those derived from the Lindbladian, we directly project the Liouville operator acting on the dynamical map, as given on the right-hand side of Eq. (94), onto the basis $\mathbf{X}$. We then subtract the resulting expression from the right-hand side of Eq. (97), which should yield zero if both formulations are equivalent. First, we define the elements of the dynamical map as in Eq. (92).

```
In[*]:= V = SparseArray[Array[v[##][t] &, {n, n}]];
```

We calculate the right-hand side of the dynamical differential equations using the algebraically derived Eq. (97).

```
In[*]:= DiffEqsAlgebraicMethod =
```

```
SparseArray[Flatten[V].Flatten[Transpose[SparseArray[ $\frac{1}{4}$  zt.Λ.z], {1, 3, 2, 4}], 1]]
```

```
Out[*]=
```

```
SparseArray[ Specified elements: 16  
Dimensions: {4, 4}]
```

Here, we project the Liouville operator obtained through the direct method, acting on the dynamical map as given on the right-hand side of Eq. (94), onto the basis $\mathbf{X}$.

```
In[*]:= DiffEqsDirectMethod = TensorContract[
```

```
Transpose[V].TensorContract[x.Transpose[x, {2, 3, 1, 4}].Ldm, {{3, 6}}, {{1, 2}}]
```

```
Out[*]=
```

```
SparseArray[ Specified elements: 16  
Dimensions: {4, 4}]
```

Finally, we compare the results obtained by both methods.

```
In[*]:= PossibleZeroQ[SparseArray[DiffEqsDirectMethod - DiffEqsAlgebraicMethod]]
```

```
Out[*]=
```

```
SparseArray[ Specified elements: 0  
Dimensions: {4, 4}  
Default: True]
```